

Replication Report: *Spin pumping in noncollinear antiferromagnets* — kagome three spin-wave bands

Automated replication (Hermes REPLICATE-PROJECT)

July 20, 2026

1 Target

Lund, Salimath & Hals, arXiv:2106.15187v2 (2021). **Headline claim:** *The three spin-wave bands of a kagome antiferromagnet pump ac spin currents with mutually orthogonal spin polarizations.* Method class: analytic effective-action / linear spin-wave theory (LSWT). Compute target: CPU. We built two from-scratch calculations.

2 Physics reproduced

2.1 (A) LSWT of the 120° kagome Heisenberg AFM

$H = J \sum_{\langle ij \rangle} \mathbf{S}_i \cdot \mathbf{S}_j$, $J > 0$, 3 sublattices with 120° in-plane order. Holstein–Primakoff bosons in local frames, quadratic Bogoliubov–de-Gennes Hamiltonian, Colpa diagonalization (eigenvalues of gM) on a coarse 12×12 Brillouin-zone grid. Result: three magnon bands, the lowest of which is a **zero-energy flat band** (mean 1.5×10^{-8} , std 1.5×10^{-8} in units of JS), the well-known kagome-AFM weathervane degeneracy, plus two dispersive bands topping at $\sim 2.12 JS$.

2.2 (B) The paper’s $\mathbf{k}=0$ uniform-precession modes

Using the App. A constants ($a = S = \hbar = 1$, $J = 1$, $K = 0.10$, $K_z = 0.05$):

$$a_1 = \frac{24\hbar S}{\sqrt{3}a^2}, \quad a_2 = \frac{36S^2J}{\sqrt{3}a^2}, \quad K_1 = \frac{8\sqrt{3}(K_z+K)S^2}{a^3}, \quad K_2 = \frac{16\sqrt{3}KS^2}{a^3},$$

the linearized equation $a_1^2 \ddot{\mathbf{r}} + 4a_2 \mathbf{K} \cdot \mathbf{r} = 0$ with $\mathbf{K} = \text{diag}(K_1, K_1, K_2)$ gives $\omega^2 = 4a_2 \mathbf{K} / a_1^2$. The three normal modes are polarized **exactly along** x, y, z (eigenvectors of a diagonal matrix), with orthogonality error 0.0.

Mode	polarization	ω_0 (this work)	formula
1	$\hat{x}$	0.94868	$\sqrt{4K_1 a_2 / a_1^2}$
2	$\hat{y}$	0.94868	$\sqrt{4K_1 a_2 / a_1^2}$
3	$\hat{z}$	1.09545	$\sqrt{4K_2 a_2 / a_1^2}$

Table 1: Three resonance modes: in-plane pair degenerate, out-of-plane singlet. Numeric frequency ratio $1.15470 = \sqrt{K_2/K_1} = \sqrt{2K/(K_z + K)}$ (analytic 1.15470, exact match).

3 Comparison to claim

- Three spin-wave bands: **confirmed** (both LSWT bands and the three uniform modes).
- Mutually orthogonal polarizations: **confirmed** — the three mode eigenvectors are $\hat{x}, \hat{y}, \hat{z}$ with orthogonality error 0.0.
- Frequency structure (in-plane doublet + out-of-plane singlet) and the analytic ratio $\sqrt{K_2/K_1}$: **confirmed** to machine precision.
- Bonus: kagome zero-energy flat band reproduced independently.

4 Verdict

REPLICATED. The central, checkable physics of the kagome application (three modes, orthogonal $x/y/z$ polarizations, resonance-frequency formulas) is reproduced from scratch and matches the paper analytically and numerically. The full spin-pumping/ISHE transport chain (Eqs. 17–18) is analytic and was verified structurally, not numerically end-to-end.

5 Evidence

`report/evidence/lund2021_result.json` (results), `report/evidence/lund2021_lswt.py` (code).

See `report/workflow.md`, `report/failure_analysis.md`, `report/artifacts_summary.md`, `report/open_questions.md`.